MILITARY SPECIFICATION

FUEL SYSTEM, AIRCRAFT, DESIGN, PERFORMANCE, INSTALLATION, TESTING, AND DATA REQUIREMENTS, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers requirements for design, performance, installation, testing and data requirements of the fuel system and fuel subsystems of aircraft power by air breathing engines. In conflicts between this document and referenced documents, this document governs.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

WW-T-700	Tube, Aluminum Alloy, Drawn, Seamless, General Specification for

Military

MIL-B-5087	Bonding, Electrical, and Lighting Protection, for Aerospace Systems
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-P-5238	Pump, Centrifugal, Fuel Booster, Aircraft, General Specification for
MIL-T-5578	Tank, Fuel, Aircraft, Self-Sealing
MIL-P-5902	Purging Gas Systems, Aircraft Fuel Tank, Internal and External, General Specification for
MIL-T-6396	Tank, Fuel, Oil, Water-Alcohol, Coolant Fluid, Aircraft, Non-Self-Sealing, Removable, Internal
MIL-L-6503	Lighting Equipment, Aircraft, General Specification for Installation of
MIL-E-7080	Electric Equipment, Aircraft, Selection and Installation of
MIL-F-7179	Finishes and Coatings, General Specification for Protection of Aircraft and Aircraft Parts
MIL-T-7378	Tank, Fuel, Aircraft, External, Auxiliary, Removable
MIL-C-7413	Couplings, Quick Disconnect, Automatic Shutoff
MIL-V-7899	Valves, Fuel Check, Low Pressure
MIL-G-7940	Gage, Liquid Quantity, Capacitor Type, Installation and Calibration of
MIL-P-8045	Plastic, Self-Sealing and Non-Self-Sealing Tank Backing Material
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Aircraft Missiles
MIL-E-8606	Tubing, Steel, Corrosion-Resistant (18-8 Stabilized)
MIL-V-8608	Valves, Fuel Shutoff, Electrical Motor Operated

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MIL-F-8615	Fuel Systems Components, General Specification for
MIL-S-8710	Strainer, Aircraft Fuel System, General Specification for
MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Pneumatic, Fuel and Oil Resistant
MIL-A-8865	Airplane, Strength and Rigidity Miscellaneous Loads
MIL-D-9402	Design of Electrical Equipment in Aircraft, General Specification for
MIL-C-25014	Coupling, Flexible, Fuel Line, 125 PSI
MIL-V-25023	Valve Fuel Drain, Self-Locking
MIL-N-25161	Nozzle, Flight Pressure Refueling, Type MA-2
MIL-H-25579	Hose Assembly, Tetrafluorothylene, High Temperature, Power Plant, Aircraft
MIL-T-25783	Tank, Fuel, Aircraft and Missile, Non-Self-Sealing, High Temperature
MIL-A-25896	Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2-1/2 Inch Diameter
MIL-S-26390	Switches, Pressure, Aircraft Fuel
MIL-G-26988	Gage, Liquid Quantity, Capacitor Type, Transistorized, General Specification for
MIL-H-27267	Hose, Tetrafluoroethylene, High Temperature, Medium Pressure
MIL-T-27422	Tank, Fuel, Crash Resistant, Aircraft
MIL-R-27521	Receptacle, Flying Boom, Aerial Refueling
MIL-V-38003	Valves, Fuel Level Control, Fuel Tank, Aircraft, General Specification for
MIL-C-38300	Connectors, Electrical, Circular, Multicontact, High Environment Quantitative Reliability, General Requirements for
MIL-C-38373	Cap, Fluid Tank Filler

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-785	Requirements for Reliability Programs (for Systems and Requirements)
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-1247	Identification of Pipe, Hose, and Tube Lines for Aircraft, Missile, and Space Systems
MS24356	Nozzle - Type MA-2, Flight Pressure Refueling
MS24359	Hose, Probe Mast - Type MA-2, Flight Pressure Refueling Nozzle, Outline Dimensions for
MS24360	Ring, Lock, Flight Pressure Refueling Nozzle
MS29520	Nozzle, Pressure Fuel Servicing Lacking, Aircraft, Type D-1
MS29525	Cap, Pressure Fuel Servicing, 2 1/2 Inch Flush Type
MS29526	Cap, Pressure Fuel Servicing, 2 1/2 Inch Diameter Non-Flush Type

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MS33514	Fitting End, Standard Dimensions for Flareless Type Connection and Gasket Seal
MS33583	Tubing End, Double Flare, Standard Dimensions for
MS33584	Tubing End, Standard Dimensions for Flared
MS33586	Metals, Definitions of Dissimilar
MS33611	Tube Bend Radii
MS33645	Receptacle, Installation, Fuel Nozzle Jumper, Aircraft to Servicing Hose

Air-Force Navy Aeronautical

AND10064	Fittings, Installation of Flared Tube, Straight Threaded Connectors

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Aviation Agency

FAA Advisory Circular 25-3	Protection of Transport Aircraft Fuel Systems Against Lightning

(Application for copies should be addressed to the Federal Aviation Agency, Washington, D.C. 20553.)

3. REQUIREMENTS

3.1 Subsystems. The fuel system shall consist of the following subsystems:

a. Engine feed and transfer subsystem

b. Aircraft fuel tank subsystem

c. Inerting subsystem

d. Pressurization subsystem

e. Fuel vent subsystem

f. Fuel quantity gaging subsystem

g. Ground fueling and defueling subsystem

h. Aerial refueling subsystem

i. Fuel jettison subsystem

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3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials used in the manufacture of fuel systems of Air Force aircraft shall be of high quality, suitable for the environment expected for the aircraft, and shall conform to applicable Government specifications or equivalent contractor specifications. The use of contractor specifications must be approved by the procuring activity. All materials shall be sufficiently resistant to applicable fluids specified in the weapon specification to assure satisfactory operation as defined herein.

3.3.1 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, atmospheric conditions or wear likely to be encountered in transportation, storage, or during normal service life.

3.3.2 Dissimilar metals. Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion.

3.3.3 Magnesium, copper, and cadmium. Magnesium, copper and cadmium plating shall not be used in parts that come into contact with fuel.

3.3.4 Protective treatment. When materials are used that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during storage or service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will chip, crack, abrade, peal, or scale with usage, age, or extremes of climatic and environmental conditions shall be prohibited.

3.4 Reliability. The reliability program for the fuel system, fuel subsystems and components shall be integrated with the overall system reliability program, and shall comply with the requirements of MIL-STD-785.

3.5 Maintainability. The maintainability program for the fuel system, fuel subsystems and components shall be integrated with the overall system maintainability program and shall comply with the requirements of MIL-STD-470.

3.6 Design and performance. The fuel system and all of its subsystems and components shall be designed in accordance with the requirements of this specification and those dictated by the operational environment and mission of the weapon system as described in the weapon system specification. The fuel system including all subsystems and components shall function satisfactorily under all possible design operating conditions for the aircraft.

3.6.1 Approval of design and development. As the development of the fuel system progresses, procuring activity engineering approval of the contractor efforts shall

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be obtained for each document that controls the design and development of the fuel system, and for each document that substantiates performance of the fuel system, its subsystems and critical components. Procuring activity approval of the installed fuel system shall be obtained before acceptance of production aircraft. Adequate documentation to define and control the development of the fuel system shall consist of all data listed in the appendix of this specification.

3.6.2 General overall fuel system

3.6.2.1 Fuel. The fuel system design shall be suitable for use of any fuel specified in the "Engine Model Specification" and "System Specification" as primary, alternate or emergency fuel. Degradation of performance with alternate or emergency fuel shall be specified.

3.6.2.2 Environmental

3.6.2.2.1 Temperature environment. The fuel system shall be designed to operate throughout the temperature range shown in table I.

3.6.2.2.2 Icing. The fuel system shall be so designed that icing does not adversely affect the fuel system components. Safety of flight of the aircraft shall not be jeopardized when operating with 0.75 cc of free water per gallon of fuel in excess of saturation at 80°F. The fuel system shall be capable of continuous operation with fuel initially saturated with water at 80°F and cooled to any temperature down to and including -65°F. The fuel system shall operate under these conditions with fuel containing no anti-icing additive without draining or cleaning of the system.

3.6.2.3 Configuration. The fuel system shall be designed for maximum simplicity consistent with weapon system requirements. The same piping and components may be utilized in more than one fuel subsystem. Fuel tanks shall not be located over crew or passenger compartments.

3.6.2.3.1 Components. Components shall comply with the applicable military specification which represents the latest technological development for that equipment. Fuel system components not covered by specific government specification shall conform to the applicable parts of MIL-F-8615 and model specifications shall be prepared and approved by the procuring activity in accordance with the appendix of MIL-F-8615. The fuel system components shall be designed for an operating pressure of 60 psi. At a burst pressure of 180 psi at maximum operating temperature there shall be no evidence of failure, malfunction, permanent distoration or external leakage. Pressure lines used in the fuel system shall have a wall thickness capable of withstanding an internal pressure of 480 psi without permanent deformation. For tanker aircraft, that part of the system used to transfer fuel to the receiver aircraft shall be designed for operating pressure of 120 psi, proof pressure of 240 psi, and burst pressure of 360 psi. All components shall be designed to be supported directly by the airframe structure and all components employing electrical devices shall be explosion proof, and shall have passed the explosion test of MIL-STD-810.

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TABLE I

TEMPERATURE ENVIRONMENT

	FUEL TEMPERATURE		AMBIENT AIR TEMPERATURE	
	LOW TEMP	HIGH TEMP	LOW TEMP	HIGH TEMP
CLASS I Fuel system which is not affected by fuel heating 1/	-65°F	+135°F	-65°F	+160°F
CLASS II Fuel system with aerodynamic heating or when fuel is used as a heat sink	-65°F	Max fuel temp as determined by Engineering Design Analysis (Item 3.a.2 of Appendix)	-65°F	Max ambient temp as determined by Engineering Design Analysis (Item 3.a.2 of Appendix)

NOTE 1/ Due to location in hot compartments, the temperature requirement for some components in a Class I fuel system may exceed the temperatures specified. These components shall be identified and the maximum temperature specified.

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3.6.2.3.2 Center of gravity control. Normal sequencing of fuel tanks shall be sufficient for maintaining center of gravity in the aircraft. It shall not be necessary to transfer fuel between fuel tanks for the express purpose of maintaining the aircraft center of gravity within limits except for supersonic aircraft.

3.6.2.3.3 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable per the requirements of MIL-I-8500. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.6.2.4 Emergency operation. The design and performance of the fuel system shall be such that no single equipment or component failure in the fuel system or the hydraulic, pneumatic or electrical system supplying power to the fuel system will cause critical structural failure or prevent safe recovery of the aircraft.

3.6.2.5 Installation

3.6.2.5.1 Fuel lines

3.6.2.5.1.1 Fuel line support and routing. Fuel lines shall be supported in accordance with table II. Rigid fuel lines shall be supported so that the lines will not deflect out of position as a result of internal pressure or as a result of aircraft maneuvers. Sufficient clearance shall be provided around fuel lines to prevent chaffing of the lines with the aircraft structure or other lines and components. Unlined metal support clamps shall not be used on metal fuel lines. All fuel lines shall be routed in the most efficient manner to obtain maximum protection against battle damage. Where possible, route fuel lines through the fuel tanks to minimize fuel losses and fire resulting from combat damage or fitting failure. Access for maintenance and inspection of line joints of the fuel system shall be provided. Fuel lines which may be subjected to maintenance damage shall be protected.

TABLE II

FUEL LINES SUPPORT DISTANCE

Tube Size Inches	Maximum Distance Between Supports Inches
1 and smaller	24
1-1/4 to 1-3/4	27
2	31
2-1/2	33
3	34-1/2
4	37-1/2
5 and over	40

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3.6.2.5.1.2 Metal tubing. Metal tubing for fuel lines shall be fabricated from either stainless steel in accordance with MIL-T-8606, or aluminum alloy in accordance with WW-T-700 both of which are weldable. Aluminum alloy tubing shall be treated in accordance with MIL-F-7179. Welded tees and wyes are acceptable in the fuel system, however, extreme caution shall be used to obtain satisfactory welds. The welded tees and wyes shall not support any external weight or equipment items unless specifically designed for this purpose. Metal tubing shall be restricted to the following diameters: 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, 1-3/4, 2, 2-1/2, 3, 4, 5, 6, and 7 inches. Hose assemblies rather than metal tubing shall be used for 1/4-inch fuel lines except for drain lines. Minimum bend radii for metal tubing shall comply with MS33611. Forcing, bending, or stretching metal tubing to accomplish installation shall not be permitted.

3.6.2.5.1.3 Hose. Hose assemblies shall be in accordance with MIL-H-8795, MIL-H-25579, or MIL-H-27267. Hose shall be installed with a bend radius no less than the minimum radius specified in the applicable military specification. Flexible hose connections when installed in the aircraft shall not be under tension, compression or shear stress.

3.6.2.5.1.4 Hot section fuel lines. Fuel lines shall not run through the hot section of the power plant compartment. If the routing of the fuel line through the hot section cannot possibly be avoided, then the line shall be stainless steel tubing with AND10064 connections. The procuring activity must approve the installation of fuel lines in the hot section.

3.6.2.5.2 Fuel fittings. Bulkhead fittings in accordance with AND10064 shall be used as connectors for all tubing, hoses, or wiring where passing through the fire wall and vapor or liquid tight bulkheads. V-Band type connectors shall not be used in fuel lines. Self-sealing type quick disconnect couplings used in the fuel system shall be in accordance with MIL-C-7413. Flexible couplings used in the fuel system shall be in accordance with MIL-C-25014. Tubing ends or fitting ends in accordance with MS33583, MS33584 or MS33514 shall be used with MS standard tube fittings in solid connections.

3.6.2.5.3 Aircraft accessories. All aircraft fuel system components shall be made accessible for inspection, cleaning, and adjustment or replacement while installed on the aircraft, with tools normally found in the mechanic's tool kit without removal of the engine, fuel tanks, or important parts of the aircraft structure.

3.6.2.5.4 Bonding. The aircraft fuel system shall be electrically bonded in accordance with MIL-B-5087.

3.6.2.5.5 Fuel drains. Fuel drains shall be incorporated in such a manner that no drainage impinges upon or re-enters the aircraft so that safe draining can be accomplished under all operating conditions. Fuel drains shall not be interconnected with component seal drains or with drain lines for electrical accessories draining oil, hydraulic fluid, water-alcohol, etc. Drain lines shall not be used for electrical

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wiring conduit. In the design of complicated nacelles and bladder tank cavities, all sheet metal pockets and traps where fuel may collect shall be drained to the exterior of the aircraft, by a sufficient number of drains, with a 3/8-inch minimum diameter tubing. Drain valves shall be in accordance with MIL-V-25023.

3.6.2.5.6 Marking. All fuel lines shall be identified by color bands in accordance with MIL-STD-1247.

3.6.2.5.7 Pipe threads. Tapered pipe threads shall not be used in the fuel system except for permanent closures.

3.6.2.6 Electrical equipment. All electrical equipment associated with the fuel system shall be in accordance with MIL-D-9402 and MIL-E-7080 except connectors shall be in accordance with MIL-C-38300. Electrical equipment shall be isolated, where possible, from the fuel system to minimize the possibility of fuel leakage and fuel vapor coming in contact with electrical equipment through either the effect of gravity, airflow, or battle damage. Wiring shall be installed in accordance with MIL-W-5088. All electrical equipment shall meet the interference requirements of MIL-STD-826.

3.6.2.7 Fire hazard reduction. Fuel lines shall not be run through any nacelle or power plant compartment except the ones they feed. The joints in fuel lines running through the cockpit shall be held to a minimum, and these joints shall be shrouded and drained to prevent fuel leakage from entering the cockpit.

3.6.2.8 Lightning hazards. The fuel system shall be designed in accordance with FAA Advisory Circular 25.3 to prevent fire and explosion when the aircraft encounters direct lightning strokes or conditions of cross field streamering.

3.6.2.9 Cockpit fuel controls. The panel for the cockpit fuel controls shall have a simplified fuel system diagram inscribed on the panel. In this diagram the fuel tanks shall be represented by their respective quantity gages and the flow control valves shall be represented by their respective control switches. The switches shall clearly indicate the ability of the fuel to either flow through the valve or be stopped by the valve. Critical switches shall incorporate safety guards. Multiple position selector switches shall not be used for fuel system controls. Dual flow controls shall be provided in tandem training aircraft with one set in each cockpit. Fuel valve controls shall be marked red and shall be grouped for convenient operation.

3.7 Engine feed and transfer subsystem. It shall be possible to feed any engine from any tank, and it shall be possible to cut off fuel flow to any engine or combination of engines without adversely affecting the fuel flow to the remaining engine(s). The fuel feed and transfer system shall be designed so that it is possible to consume the fuel in the main tanks without depending on boost pumps at altitudes up to 10,000 feet above sea level for fuel flow rates necessary for maximum engine power settings. Fuel from every tank shall be made available to the engines by suction feed at a fuel flow rate corresponding to cruise condition. External tanks are excluded from the

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suction feed requirements except where they serve as main tanks. Fuel from jettisonable external tanks with a capacity less than 1000 gallons shall be transferred by pressurization of the tanks. For nonjettisonable tanks and tanks with capacity greater than 1000 gallons either may be used. The fuel system shall be capable of satisfactory performance during maximum rate of climb to the aircraft service ceiling, with each engine operating from its normal fuel supply, and with fuel in each tank at a temperature of 110°F at the start of the climb. This same performance is required with the fuel being pumped directly from the auxiliary tanks to the engine.

3.7.1 Engine inlet fuel feed conditions

3.7.1.1 Normal operation. The condition of the fuel with regard to pressures, temperature, and all parameters which may affect engine operation, shall satisfy the conditions listed in the engine model specification under the most adverse conditions. The fuel system shall in no way limit aircraft performance within allowable engine and aircraft operating limits.

3.7.1.2 Boost pumps inoperative. The V/L ratio of the fuel at the fuel inlet to the engine shall not exceed the V/L ratio specified by the engine model specification at altitudes up to 10,000 feet at military rated power with fuel boost pumps inoperative. This condition is determined for the worst possible operational condition of the feed system; that is, with the airframe fuel filter in the impending bypass condition at 110°F or the maximum fuel design temperature whichever is higher.

3.7.2 Transfer system arrangement. The transfer system shall be so designed that the immediate attention or immediate action of the pilot (or flight engineer) during normal fuel system operation is not required for safety of flight. Transfer pumps shall be shutoff automatically when the tank is empty on aircraft where the fuel system controls are not located on the flight engineer's panel. The transfer system shall deliver fuel to the main tank(s) or directly to the engines by an override system, at a rate not less than the rate of fuel consumption of the engine(s) in the military power condition for all altitudes.

3.7.3 Fuel availability warning. A low fuel level warning shall be provided in all aircraft. The low level warning should indicate 20 minutes of fuel remaining at maximum range cruise power and altitude plus fuel for a normal descent and landing plus one missed approach.

3.7.4 Unavailable fuel. Unavailable fuel shall be held to a minimum consistent with sump capacity requirements. Tank outlets and boost pumps shall be so located that maximum main tank fuel is usable in landing approach and maximum auxiliary tank fuel is usable during level flight. Ejectors may be utilized to reduce the amount of unavailable fuel.

3.7.5 Extreme operating attitude. The aircraft shall be capable of uninterrupted fuel feed for a period of 3 minutes in any extreme operating attitude excluding inverted flight, with the main tank(s) 25 percent full. The airframe manufacturer shall specify

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the maximum allowable extreme attitude for pitch, yaw, and roll. A full continuous fuel flow shall be provided for the tank(s) to the engine(s) for at least 10 seconds during negative "G" for fighters, trainers, or any other aircraft required to fly in this manner.

3.7.6 Functional failure

3.7.6.1 Take-off with failure. With any single failure of a functional component of the feed and transfer system, it shall be possible to attain 10,000 feet altitude and execute a safe return to base.

3.7.6.2 Failure during flight. No single failure of a functional component in the fuel system or other system necessary to the proper functioning of the fuel system shall result in an unsafe condition.

3.7.7 Feed and transfer surge requirement. Surge pressure shall not exceed 120 psi as a result of initiating or stopping inter-tank transfer or as a result of decreasing power settings of the engine(s).

3.7.8 Feed and transfer subsystem components

3.7.8.1 Fuel boost pumps. Boost pumps shall conform to the requirements of MIL-P-5238. The boost pumps installed in fuel tanks of 1000 gallons or greater capacity shall be removable without draining the tank. For pumps with three phase ac motors which are installed for two-phase operation, the two-phase operation must be defined in the pump model specification and approved by the procuring activity.

3.7.8.2 Fuel pressure switches. Each engine feed line shall contain a fuel pressure switch. Fuel pressure switches shall conform with the requirements of MIL-S-26390. Low fuel pressure warning lights shall be installed in the cockpit on the fuel control panel.

3.7.8.3 Airframe fuel strainers. The necessary strainers or filters shall be incorporated in the airframe fuel system to insure that the particle size of contaminants delivered to the engine is within the limits set forth in the applicable engine model specification. Airframe fuel strainers shall be in accordance with MIL-S-8710. The strainer shall be so located with respect to the fuel tanks and lines that the entire fuel system may be drained by opening the strainer drain, fuel tank sump drain, and booster pump drain plug. All fuel system strainers shall be equipped with bypass provisions to prevent engine fuel starvation by icing or plugging of the strainer element with contaminant. The bypass shall be designed to minimize washing ice or contaminant from the element or sump into the fuel system. An indicator shall be provided to show when the strainer is impending bypass. All fuel strainers shall be installed to be readily accessible, properly mounted by means of the mounting lugs provided, and with sufficient clearance to permit easy removal of the lower cover and element.

3.7.8.4 Fuel shutoff valves. Electrically operated shutoff valves shall conform to the requirements of MIL-V-8608. Electrically operated valves shall be so designed

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that there is no possibility of the valve changing position during flight due to a short in the aircraft electrical circuitry supplying power to the valve, to electrical power failure, or normal mechanical forces. The power line for each critical engine shutoff valve shall be connected to the essential bus with a minimum number of electrical components, such as plugs and relays, in the circuit. The critical valve wires shall be shielded when contained in wire bundles. The shutoff valves shall be located as near the engine compartment as possible, but not in the compartment. The preferable location is on the outside face of the engine compartment firewall. Cockpit warning lights shall be provided for the firewall shutoff valves and fuel dump valves that will illuminate when the valve is in transition, a fault exists on the valve circuitry, or the valve limit switch fails to operate.

3.7.8.5 Check valve. Check valves shall be in accordance with MIL-V-7899.

3.8 Aircraft fuel tank subsystem. The fuel tanks on all aircraft shall be designed in conjunction with other fuel system components so that the fuel is continuously available to the engine under all design maneuvers of the aircraft. All fuel tanks shall be completely drainable while the aircraft is at ground attitude. The type of tankage shall be selected from table III after careful consideration of the aircraft mission relative to reliability, maintainability, passive defense features, and crash safety aspects.

3.8.1 Internal tanks. The initial layout of the aircraft shall determine the most advantageous shape of the fuel tank. Shallow or flat horizontal tanks with flat bottoms shall not be used except for thin winged aircraft in which shallow fuel tanks are necessary.

3.8.1.1 Tank accessibility. Fuel tanks shall be designed to be accessible. All tanks except integral tanks shall be installed so that it is possible to remove the tanks without:

a. Removing any other part of the aircraft, except cowling or access panels

b. Disassembling structural parts

c. Removing the engine or any part of the engine.

Ample clearance shall be provided for the removal of sumps or inspection plates.

3.8.1.2 Tank construction. Internal structure within the tank, such as baffles, flapper valves, and vent lines shall be designed and constructed to be as simple and reliable as possible. The very minimum of such internal devices shall be utilized to comply with the fuel system performance requirements. The number of cells and the number of fittings required per cell shall be held to a minimum. The fittings shall be located above the fuel level wherever possible without any sacrifice of performance. "O" rings or "mold in groove" seals shall be used for sealing openings. Tank design and construction shall be in accordance with the applicable specification of table III.

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TABLE III

SELECTION OF PROPULSION FLUID TANKS FOR AIRCRAFT

Type of Tankage	Design Criteria Document	Applications and Restrictions
Conventional Bladder Cells	MIL-T-6396	Select conventional bladder cells for application where protection against small arms or maximum crash resistance are not primary objectives and where fuel temperatures do not exceed 135°F or cell wall temperatures do not exceed 165°F.
Self-Sealing	MIL-T-5578	Utilize self-sealing tanks where protection against gunfire up to 20 MM is a requirement.
Non-Integral, Internal Non-Flexible Tanks	MIL-T-6396	Use for special applications subject to approval of the procuring activity.
Crash Resistant Tanks	MIL-T-27422	Utilize crash resistant tanks for fuselage and wing tanks when primary mission of aircraft is personnel transport.
High Temperature Tanks	MIL-T-25783	Utilize high temperature tanks in high performance aircraft when fuel temperature exceeds 135°F or when cell wall temperatures exceed 165°F.
External Tanks	MIL-T-7378	Use for range extension or to meet ferry requirements.
Integral Tanks	- - -	Use only for wing tanks and after careful consideration of other tanks listed above. Do not use integral tanks for center wing or fuselage installation or directly adjacent engine or gun compartments on manned aircraft. Do not select integral tankage for aircraft of which the primary mission is ground attack, low altitude observation, or other missions which subject the aircraft to small arms fire. Before selecting integral tanks, consider carefully the problems of structural corrosion due to water collection and microorganism growth.

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3.8.1.3 Expansion space. An expansion space to prevent tank overflow shall be provided in each of the fuel tanks equal to or greater than three percent of the total fuel volume of the tank with the aircraft in ground refueling attitude. Where two or more tanks are interconnected to serve as a single tank, expansion space may be provided for the tank cluster instead of for the individual tanks.

3.8.1.4 Sumps. Each fuel tank shall be provided with a sump (which shall be the lowest portion of the tank when the aircraft is at ground attitude) for the purpose of collecting sediment and water.

3.8.1.5 Sump capacity. The sump shall have a capacity equivalent to 0.0025 gallons per gallon (.25 percent) of tank volume, to contain the water released from solution with the fuel in addition to that condensed in the fuel cells. The capacity of the sump shall be never less than 1 pint.

3.8.1.6 Sump drains. Sump drain valves shall be in accordance with MIL-V-25023. Drain bosses provided on booster pumps shall not be used as fuel tank sump drains.

3.8.1.7 Access doors. Bolts smaller than 1/4 inch diameter shall not be used for access doors or for installing components on the tank. Each cell shall be provided with an access door of such size that the entire interior of the tank is available for inspection and cleaning.

3.8.1.8 Outlet. The fuel tank outlet shall be located in such a position that water cannot collect and freeze around the impeller of the booster pump in cold weather. The fuel tank outlet fitting shall be so located as to reduce the unusable fuel to a minimum compatible with the sump capacity requirements of 3.8.1.5.

3.8.1.9 Tank accessories. Tank accessories, such as pumps, sumps, gages, vent lines, and outlet lines shall be mounted so that their load is carried by aircraft structure.

3.8.1.10 Filler openings. The filler opening shall be so located to make it impossible to fill the expansion space. The filler openings shall be located to permit filling the tanks from outside the aircraft when it is on the ground without the use of special funnels. Filler openings shall also be located so that the liquids from the openings do not spill on parts of the aircraft structure, engine, or accessories, and thus cause damage or a hazardous condition. If necessary, filler spill basins shall be provided with adequate drains to prevent spilled liquid from entering the fuselage, engine compartment, and wings. The tank shall be properly vented so that when the filler cap is removed, complete filling can be accomplished in only one operation. Screens shall not be located in the filler neck. An internal guard shall be provided for thin wing aircraft having shallow fuel tanks to prevent the fuel nozzle from striking and damaging the bottom of the tank. Such a guard shall not reduce the required filling rate. Filler caps shall be in accordance with MIL-C-38373.

3.8.1.11 Tank identification. Removable tanks shall be marked for identification in accordance with the applicable specification of table III.

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3.8.1.12 Installation

3.8.1.12.1 Tank installation time. The time required to completely remove and install a new flexible internal tank (self-sealing or non-self-sealing) fuel tank in an aircraft of four or more engines shall be 8 hours maximum; on aircraft with less than four engines the time shall be 4 hours maximum.

3.8.1.12.2 Rigid internal removable tanks. Rigid internal removable fuel tanks shall be supported in padded cradles or on padded supports suitable for distributing the weight equally throughout the tank structure. Metal fuel tanks shall not be mounted rigidly to the aircraft. The padding shall consist of gasoline and oil resistant material which will not absorb liquids. Felt-type padding shall not be used. Means shall be provided for holding the tank in place under adverse loading conditions. Brackets or lugs attached to the tank walls shall not be used for supporting the tank.

3.8.1.12.3 Self-sealing tanks. Metallic fuel lines shall not be installed in contact with the walls of a self-sealing tank. Aircraft structure surrounding or supporting self-sealing tanks shall be designed and constructed to withstand all flight conditions and forces produced by the passage of a projectile through the confined liquid. All external self-sealing surfaces of a flexible self-sealing tank shall be protected by using backing board conforming to MIL-P-8045.

3.8.1.12.4 Bladder tanks. Non-self-sealing bladder tanks shall be installed in cavities free from sharp projections or abrasive surfaces. Cavities will be free of all metal chips or other foreign material that might damage the bladder.

3.8.1.12.5 Integral tanks. Integral fuel tanks shall be designed and tested in accordance with applicable military specification.

3.8.2 External tanks

3.8.2.1 Construction. The external fuel tanks shall be constructed in accordance with MIL-T-7378. The external fuel tank shall be designed for simplicity, low cost, and ease of mass production. For jettisonable tanks fuel and vent valves shall be located in the aircraft, not in the tanks. See 3.7, 3.11.7 and 3.12 for transfer, vent, and fuel gaging requirements. The tank shall be designed so that loads are carried primarily in the tank skin and utilize a minimum number of structural bulkheads. Lap joints or continuous seam welding shall not be used in joining segments of tank skins. Spot or seam welds shall not be used to attach seal rings or bulkheads to the tank skin. Sealing compounds shall not be used. Rivets shall not be used through the tank skin, access doors, or other tank components where sealing is required.

3.8.2.2 Installation. The external fuel tanks shall be designed and installed in accordance with MIL-T-7378. The tanks installation shall provide for the following:

a. A mechanical and electrical release, when jettisoning is required

b. Approved moistureproof electrical connection

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c. Approved self-sealing quick disconnects for fuel and air pressure lines

d. Hinges and quick release type locking provisions for all pylon doors and access covers necessary for tank installation or removal.

3.8.2.2.1 Ground clearance. Design external tanks so that there will be a minimum of 3 inches clearance on a smooth runway or, if the aircraft is designed for rough runway operation, a minimum of 6 inches should be provided with strut compressed and one tire flat for either landing or takeoff configuration combined with a corrugated landing strip.

3.8.2.2.2 Installation time. The time required to install the external tank shall not exceed 15 minutes.

3.9 Pressurization, purging and inerting subsystems

3.9.1 Pressurization. The pressurization subsystem shall provide for pressure relief to prevent over pressurization of the tank as a result of a pressure regulator failure. The system shall provide for automatic venting to atmosphere whenever the pressurizing source is inoperative or shut off.

3.9.2 Purging. Purging systems installed shall be in accordance with MIL-P-5902.

3.9.3 Inerting. A fuel inerting system shall be provided for aircraft that operate in a speed region where the fuel tank wall temperatures exceed the spontaneous ignition temperature of the fuel.

3.9.3.1 Capacity. The system capacity shall be adequate to supply inerting capability for the maximum duration of flight including 20 percent margin for safety.

3.9.3.2 Ullage. At no time during the flight shall the oxygen in the ullage volume exceed 5 percent 0_2 by weight in a gas inerting type system.

3.10 Vent subsystems. The subsystem shall be designed to prevent siphoning and spillage of fuel and to protect the fuel tanks from destructive pressures. Fuel spillage through the vent line during any maneuver shall not exceed 0.05 percent of the fuel from any one tank or 1 gallon, whichever is smaller.

3.10.1 Vent line routing. With the aircraft at ground attitude, the vent line shall be routed from the highest point within the tank, transversing the three dimensions of the tank and terminating in a suitable location outside of the aircraft. Siphon breakers shall be installed in the vent lines if necessary to prevent siphoning.

3.10.2 Vent line size. The vent line size shall be selected by the length of the vent line and the amount of air or vapor that must be handled. The volume of air or vapor that must be handled shall depend on the fuel consumption of the engine, the size of the fuel tank, the rate of refueling, the type of fuel being used, the rate of climb,

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the rate of dive, and the service ceiling of the aircraft. The vent lines shall be of adequate size to prevent tank overpressurization in the event of a level control valve failure during refueling. For refueling assume that each tank is filled individually with a refueling static pressure of 55 psi at the inlet to the refueling nozzle. Diameter of vent lines shall be a minimum of 1/2 inch.

3.10.3 Vent outlet location. The tank vent outlet shall be located so that liquids and vapors emitted will not blow back into the aircraft or come in contact with the exhaust system, and located so that foreign matter thrown up by the wheels of the aircraft will not enter or block the vent outlet. The outlet shall be located so that any moisture collecting in the line will drain to the outlet and will not permit the entrance of water into the vent system. The vent outlet location and configuration shall prevent the formation of ice which will block the outlet.

3.10.4 Interconnected vents. The fuel tank vent systems shall be designed so that fuel connot be transferred from one tank to another unless those tanks supply fuel as a single unit. Fuel tank vents may be terminated at the same outlet location in the aircraft.

3.10.5 Vent pressures. For flexible non-metallic and light-walled metallic fuel cells, the vent outlet terminal shall be designed and located so that the pressure in the vented space of the fuel tanks is never below ambient pressure. For bladder type tanks, the cell cavity pressure shall never be greater than the pressure within the tank. Internal fuel tank with structural metal wall may have negative pressure within structural limits of the tank.

3.10.6 External tank vents

3.10.6.1 Vent installation. An approved fuel and pressure vent system with self-sealing quick disconnects shall be installed for all jettisonable fuel tanks. The tank shall be capable of automatic venting to atmosphere whenever the pressurizing source is shut off, and this venting equipment shall be provided in the aircraft, not in the tank.

3.11 Fuel quantity gaging subsystem. All of the internal and external fuel shall be gaged.

3.11.1 Components. All of the components of the fuel quantity gaging subsystem shall meet the requirements of MIL-G-26988.

3.11.2 Installation. The fuel quantity gaging subsystem shall be installed in accordance with the requirements of MIL-G-7940.

3.11.3 Contractor specification. In the event that the requirements of MIL-G-26988 and MIL-G-7940 are not stringent enough or do not adequately define peculiar installation requirements, the contractor may prepare a component or installation specification as required. This specification shall clearly indicate the differences between it and

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MIL-G-26988 and MIL-G-7940. In no case shall a contractor specification be prepared wherein less stringent requirements are quoted. The contractor specification shall be approved by the procuring activity.

3.12 Ground refueling and defueling subsystem. It shall be possible to refuel the aircraft to capacity without the application of external power. Provisions shall be incorporated for the capability of selecting any individual fuel tank for filling and conversely to avoid filling any tank. The fuel line between the main tank(s) and engine(s) shall not be subjected to the pressures imposed on the refueling manifold during refueling.

3.12.1 Pressure refueling. All aircraft with an internal fuel capacity of 600 gallons or more will incorporate a pressure refueling system.

3.12.1.1 Fuel servicing adapter. Fuel servicing adapters in accordance with MIL-A-25896 shall be provided for refueling the complete aircraft. The number of adapters shall be in accordance with the requirements of table IV. These adapters shall be located on the aircraft so that the use of ladders, supports or elevating devices are not required for insertion of the nozzle.

3.12.1.1.1 Adapter grounding. All fuel servicing adapters shall be positively grounded to the airframe. An electrical ground receptacle shall be installed for grounding the fuel servicing nozzle. The ground receptacle shall be in accordance with MS33645 except that the receptacle shall be located:

a. Not more than 20 inches or less than 5 inches from the servicing adapter

b. No closer than 12 inches from fuel tank vents or openings.

3.12.1.1.2 Adapter cap. Fuel servicing adapters shall be provided with MS29525 caps for flush type installations or MS29526 caps for non-flush type installations.

3.12.1.1.3 Adapter installation. The adapter shall be installed to accommodate an MS29520 ground servicing nozzle. Design the adapter installation so that the nozzle has ample clearance for installation and operation by personnel wearing heavy gloves. There shall be no switch or valve actuators installed on the adapter which will not completely clear the nozzle envelope when the nozzle is connected.

3.12.1.2 Refueling and defueling lines. Fuel lines shall be sized so that all tanks theoretically reach the full position simultaneously when refueling. For aircraft incorporating both ground and aerial refueling subsystems, the refueling line sizes shall satisfy the flow requirements of both systems. The line sizes shall be capable of handling flow schedules for pressure refueling as specified in table IV. The following criteria shall be used when designing to meet the ground refueling schedules of table IV:

a. It shall be assumed that the refueling pumping system is capable of delivering any flow up to 600 gpm for each servicing adapter at pressures of 50 ±5 psi measured at the inlet of the refueling nozzle.

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TABLE IV

PRESSURE REFUELING TIME

TOTAL AIRCRAFT FUEL WEIGHT (POUNDS)	GROUND REFUELING		MAXIMUM AERIAL REFUELING TIME* (MINUTES) (Time may be interpolated between divisions)
	MAXIMUM NUMBER OF SERVICING ADAPTERS PERMITTED	MAXIMUM TIME* (Minutes)	
450,000	4	40	50
350,000	4	32	40
250,000	2	32	30
100,000	2	20	15
50,000	1	18	10
25,000	1	8	7
10,000	1	5	4
3,900	1	2.25	2.25

*Based on refueling 90 percent total aircraft fuel (See 3.12.1.2b)

b. It shall be assumed that all tanks except main tanks are empty and that 10 percent of the total aircraft fuel capacity is distributed equally in the main tank(s) at the start of the refueling operation.

All refueling lines shall have thermal relief provisions. All fuel trapped in the refueling manifold that cannot be used for engine feed shall be scavenged or drained.

3.12.1.3 Fuel level control valves. Fuel level control valves shall be designed in accordance with MIL-V-38003. The valves shall be single or dual seat valves with dual pilot control devices, except for jettisonable external tanks, where only one control device is required. These pilot control devices shall be level sensing and not operated by sensing tank pressure. The valves shall be designed to prevent surge pressure in the refueling manifold from exceeding 120 psi. The design shall also provide for checking the operation of the primary and secondary pilot control devices in each tank, in the initial portion of the ground refueling operation.

3.12.2 Gravity refueling. It shall be possible to gravity refuel the aircraft in 4 minutes or at a minimum rate of 200 gpm. It shall not be necessary to perform any operations other than removing the fuel filler cap prior to gravity refueling the system.

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3.12.3 Ground defueling. It shall be possible to completely defuel the aircraft through the servicing adapter at a minimum rate of 200 gpm per adapter with a suction pressure at the adapter not to exceed 3 psi. The aircraft fuel boost pumps may be utilized for defueling.

3.13 Aerial refueling subsystem

3.13.1 Receptacle-flying boom. The following requirements apply to production or retrofit installation of boom/receptacle type aerial refueling system. The receptacle with slipway shall be located forward of the windshield to provide an adequate view of the refueling operation by the pilot or co-pilot. The receptacle face shall be at least 2 feet from the windshield.

3.13.1.1 Angle of receptacle installation. A composite layout shall be prepared to determine the angle at which the receptacle is to be inclined on the receiver aircraft by referencing all normal changes in the receiver's fuselage pitch angle relative to the tanker airplane's centerline with refueling hardware in the mated configuration. With smallest receiver fuselage pitch angle at the tanker booms' lower, outer limit (40° elevation, 551 inches extension) as one extreme and the largest receiver pitch angle at the tanker booms' upper, inner limit (20° elevation and 404 inches extension), the angle of installation shall be determined by the following equation:

$$X = 1/2\ (60° - A - B)$$

Where A = minimum fuselage pitch angle during refueling

B = maximum fuselage pitch angle during refueling

X = angle of installation.

The receptacle installation angle as determined by the equation may be slightly adjusted to provide optimum clearance of the tanker boom nozzle with the aircraft structure surrounding the receptacle. The composite layout as required above, the angle calculation, and any other information necessary to determine the receptacle angle of installation shall be approved by the procuring activity.

3.13.1.2 Receiver/tanker performance comparison. A performance comparison of aerial refueling speeds and altitude envelopes of tanker and receiver shall be required. The receiver shall be able to close on the tanker for refueling hookups in all refueling speed ranges without abrupt engine thrust changes such as would be produced by an unmodulated afterburner. Prior to testing, data or calculations illustrating performance compatibility shall be approved by the procuring activity.

3.13.1.3 Aerial refueling components. All receptacle installations shall incorporate the components and installation features as follows:

3.13.1.3.1 Receptacle. The receptacle, including the induction coil and electrical harness and connector, nozzle latches, mounting and trunnion support, toggle contact

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switch, toggle latch switch, hydraulic actuator, fuel valve, hydraulic toggle latch actuator, and hydraulic relief valve shall be designed and tested in accordance with MIL-R-27521. The support structure or mounting provision in the aircraft shall be designed to accept the loads as specified in the receptacle specification. The receptacle shall be located so that the entire boom envelope may be used and so that fuel spray on disconnect will not enter any air intake ducts. The receptacle and slipway shall be located on the centerline of the aircraft. Avoid locating antennas or any other aircraft protrusions in the vicinity of the receptacle and boom envelope. Receptacle/slipway shall be hydraulically actuated with fail-open provisions for emergency refueling. The receptacle location and installation features shall be approved by the procuring activity before installation of the receptacle.

3.13.1.3.2 Slipway. A slipway shall be provided to assist the boom operator in inserting the boom. The slipway must be a minimum of 18 inches long, and the sides of the slipway should be smooth and high enough to guide the nozzle into the receptacle. The nozzle radius of 2 1/2 inches shall be used as a minimum depth for guiding the nozzle into the receptacle. The slipway shall mate with the receptacle and provide smooth transition surfaces. The slipway area shall clear the boom with the maximum receiver angle of attack by a minimum of 5 degrees for any boom angle within the tanker boom automatic disconnect envelope. Slipway design for fighter aircraft may use 2 degrees clearance. The slipway shall have hydraulically operated door(s). The slipway doors shall be designed to fail in the open position for emergency refueling. Cockpit indication light shall be provided to show when the receptacle/slipway is opened and locked, and closed and locked. The design of the slipway shall be approved by the procuring activity prior to installation.

3.13.1.3.3 Night refueling lights. The receptacle/slipway installation shall incorporate two flush lights, one on each side of the slipway for night refueling. The lights shall be directed toward the receptacle bore, and large light angles shall be avoided to prevent boom operator or receiver pilot blinding. The prism lens feature may be utilized. The lights shall be designed in accordance with MIL-L-6503 and shall be capable of being dimmed from the cockpit from full bright to off.

3.13.1.3.4 Indicator lights. Indicator lights shall be provided to inform the receiver pilot of the refueling condition. The lights should read from top to bottom as follows:

Blue	"Ready" for contact
Green	"Latched"
Amber	"Disconnect"

The lights shall incorporate sun shields, transparent lens with translucent centers, press-to-test switches and a rheostat. The lights shall be located where the pilot and co-pilot may easily observe without changing visual tanker aircraft reference. The lights shall be so wired that illumination occurs only upon termination of the function being indicated. The indicator lights shall be designed in accordance with MIL-L-6503.

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3.13.1.3.5 Amplifier, reset and manual override switch. A reset switch shall be provided for returning the amplifier to precontact position (ready for contact). The reset switch shall be located near the indicator lights or at a position easily accessible to the pilot without changing visual tanker reference. A manual override switch shall be furnished for the signal amplifier to permit operation of the latches for connect and disconnect in the event of an amplifier failure. Locate the override switch for easy operation without changing the receiver pilot visual tanker reference.

3.13.1.3.6 Disconnect switch. A disconnect switch shall be provided in the aircraft control stick grip for the pilot and co-pilot. If the co-pilot position is such that the refueling cannot be conducted from this position, the co-pilot disconnect switch shall be deleted.

3.13.1.3.7 Fuel pressure disconnect switch. A switch for initiating disconnects when manifold pressures indicate a full tank condition or that high level fuel valves have all closed shall be provided. The switch may be installed in the fuel line near the receptacle and the settings shall be compatible with high level control valves; that is, a valve which will not create inadvertent disconnects.

3.13.1.3.8 Slipway and receptacle drains. Slipway and receptacle drains shall be provided to prevent accumulation of fuel spillage or water in the receptacle/slipway area. Overboard drains shall be located so that drained fuel and water will not enter any air intake ducts.

3.13.1.4 Retrofit installations. For retrofit installations of aerial refueling provisions on aircraft which do not have fuel level control valves in accordance with 3.12.1.3, the installation of the proper level control valves shall be included in the retrofit.

3.13.1.5 Reverse refueling. For aircraft with reverse refueling mission requirement, the procuring activity shall be consulted for the specific design requirements for the aircraft.

3.13.2 Probe drogue. The probe mast and actuating mechanism shall be designed and tested in accordance with MIL-F-8615. The probe mast shall be free of protruding or sharp objects which could catch or damage the drogue. A composite layout shall be prepared to determine the angle at which the probe is inclined on the receiver aircraft by referencing all normal changes in the receiver pitch angle relative to the tanker airplane centerline with refueling hardware in the mated configuration. The composite layout, the angle calculations, and any other information necessary to determine the probe angle of installation must be approved by the procuring activity.

3.13.2.1 Probe loads for the boom to drogue adapter kit refueling. The probe mast and attaching structure shall be designed for ultimate loads of 1000 pounds tension force in combination with a 3000-pound radial force, and a 2000-pound compression force acting separately as specified in MIL-A-8865.

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3.13.2.2 Probe loads for reel system refueling. The probe mast and attaching structure shall be designed for ultimate loads of 1000 pounds tension force in combination with a 1000-pound radial force, and a 2000-pound compression force acting separately as specified in MIL-A-8865.

3.13.2.3 Probe standard parts. The following standard items shall be used on the probe:

a. The probe mast nose shall be in accordance with MS24359

b. The lock ring shall be in accordance with MS24360-2

c. The MA-2 nozzle shall be in accordance with MIL-N-25161 and MS24356.

3.13.2.4 Probe location, envelope and alignment. The probe shall be located so that the receiver pilot can easily observe the refueling operation without changing his visual tanker aircraft reference. The pilot shall see a minimum of 30 inches of the probe mast measured from the tip of the nozzle. The probe shall have a minimum clearance with respect to the fuselage in accordance with figure I. The nozzle vertical axis plane shall be parallel to the vertical axis plane of the aircraft. The longitudinal axis of the nozzle shall be parallel to the line of average fuselage pitch angle. The line of average fuselage pitch angle is a line bisecting the angle between the minimum fuselage pitch angle and maximum fuselage pitch angle during the refueling operation.

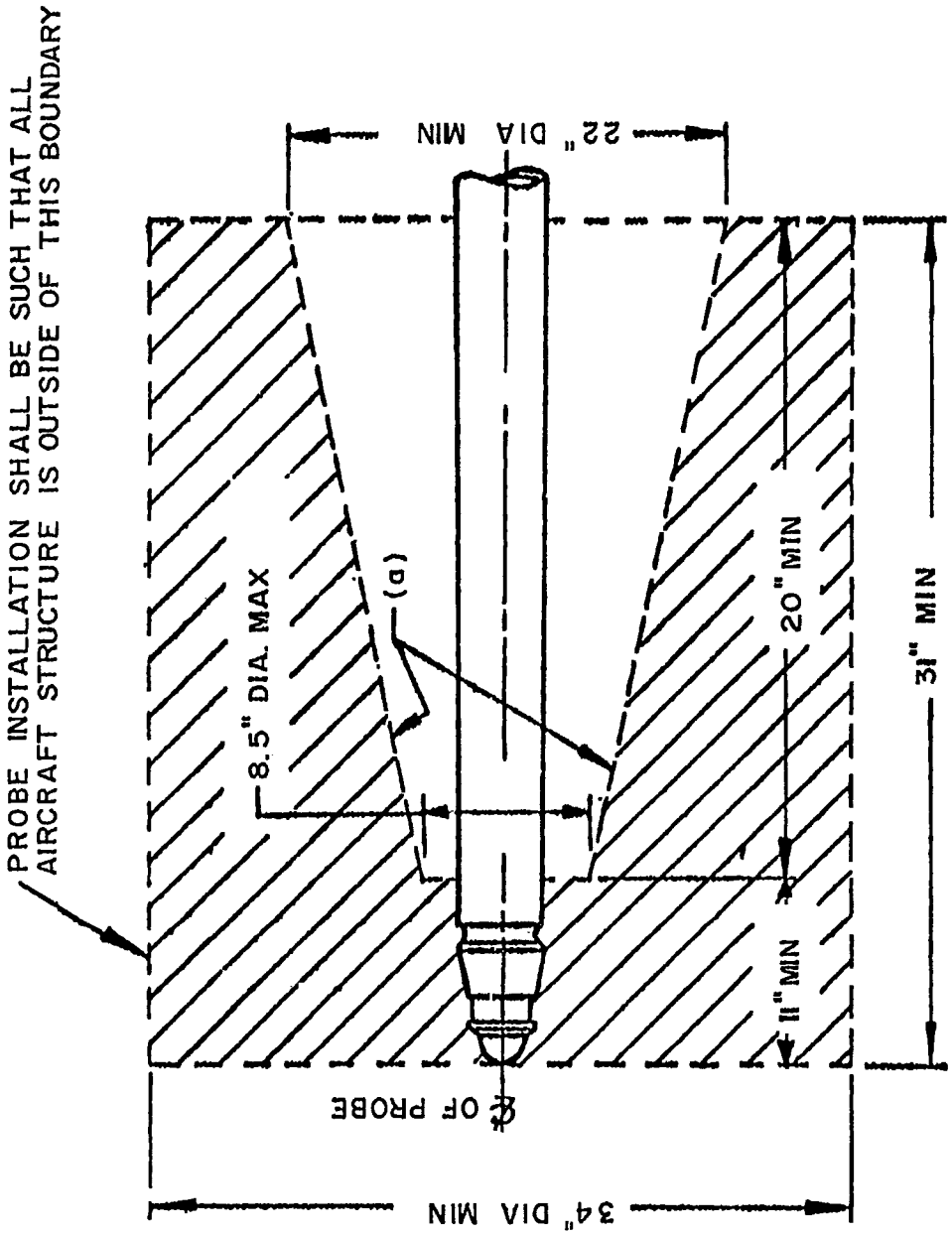
3.13.2.5 Externally mounted probe. Any probe mast which cannot be retracted into the fuselage for a flush installation shall be easily removable on the ground. The probe mast shall have the capability of being removed within 15 minutes by a maximum of two men. A drain plug shall be provided for draining the probe mast on removal. Lockwiring of the drain plug shall be provided on installation. The aircraft fuel tube inlet and probe mast outlet shall be sealed on removal of the probe mast.

3.13.2.6 Retractable probe mast. Normal extension and retraction shall be accomplished by a hydraulic actuator. Emergency backup shall be provided for extending the probe at least one time per flight. The probe stowage compartment shall be sealed to prevent fuel leakage into other aircraft compartments. The stowage compartment shall be provided with drains in accordance with 3.6.2.5.5 to prevent accumulations of fuel and water.

3.13.2.7 Telescoping probe mast. The telescoping probe shall have positive mechanical locking and pilot indication of the full extension mode and the fully retracted mode.

3.13.2.8 Lighting. All probe installations shall be provided with light(s) in accordance with MIL-L-6503 for night refueling. The light shall be capable of being dimmed from the cockpit, from full bright to off. Prior to contact the light(s) shall illuminate a minimum of 30 inches of the probe measured from the nozzle tip and the drogue at a distance of 50 yards. After hookup is established the drogue and a minimum of 9 feet of hose shall be illuminated sufficient for pilot monitoring.

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(a.) When no other probe mast or aircraft location is available for mounting probe doors, lights, etc., the envelope boundary specified shall be maintained.

FIGURE I. Probe clearance envelope

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3.13.3 Aerial refueling hardware isolation. A check valve or motor operated shutoff valve shall be provided to prevent aircraft system pressure on either the boom receptacle or the probe nozzle. When motor operated valves are employed, they shall incorporate a manual override feature, and be accessible during flight.

3.14 Fuel jettison subsystem. A fuel jettison subsystem shall be provided to provide for inflight reduction of gross weight.

3.14.1 Fuel jettison rate. The fuel jettison system shall provide a minimum jettison rate per minute equivalent to 1 percent of maximum gross weight of the aircraft, and the aircraft shall be able to jettison fuel to a safe landing weight in a maximum of 10 minutes.

3.14.2 Aircraft controllability. It shall be insured that the fuel jettison operation does not adversely affect the controllability of the aircraft.

3.14.3 Jettison controls. Means shall be provided for closing the jettison valves during flight at any stage of the jettison operation. The fuel jettison control shall be located at the systems engineer's station. In the absence of a systems engineer's station, the control shall be made accessible to the pilot or co-pilot. Design the control to minimize the possibility of the jettison system being inadvertently or accidentally actuated. The design of the jettison system shall be such that the entire fuel load cannot be dumped unless a second or additional initiating action is taken by the operator.

3.14.4 Fire hazard. The fuel jettison system and its operation shall be free from fire hazard. The fuel shall discharge clear of all parts of the aircraft. Insure that no fuel fumes enter any portion of the aircraft.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. The aircraft fuel system shall be inspected at the mock-up inspection and the inspection of the first complete experimental aircraft by engineering representatives of the procuring activity. The purpose of these inspections shall be to assist in determining compliance with the following requirements of this specification:

a. Design

b. Configuration

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c. Installation

d. Fire hazard

e. Any other requirements for which visual inspection will be beneficial in supplementing the contractor's engineering and test data.

4.3 Phases of testing. The testing of the fuel system shall be conducted in the following phases:

a. Component

b. Fuel system rig testing

c. Ground and flight testing

4.4 Component testing. Components shall be tested in accordance with the applicable military or contractor specification. Contractor specifications must be approved by the procuring activity prior to testing.

4.5 Fuel system rig testing. The fuel system rig testing shall be conducted in accordance with all the tests specified in 4.8, Rig Test Methods. With approval of the procuring activity, tests may be interchanged between 4.8 and 4.9.

4.6 Ground and flight testing. The ground and flight testing shall be conducted in accordance with section 4.9.

4.7 Test conditions

4.7.1 Test fluid. Unless otherwise specified, and approved by the procuring activity, one of the fluids specified in 3.6.2.1 shall be used.

4.7.2 Test rig. The test rig shall be capable of demonstrating fuel system performance under all operational altitudes and attitude of the aircraft. The full scale aircraft fuel system shall be simulated by the test rig. Where practicable, the test rig shall incorporate the actual fuel system components (pumps, valves, piping, connections, etc.) that are intended to be utilized in the aircraft fuel system installation. Prior to start of testing, the complete test outline and description of the test rig shall be submitted to the procuring activity for approval.

4.8 Rig test methods

4.8.1 Engine feed and transfer subsystem tests

4.8.1.1 Fuel availability

a. Rig tests shall be conducted to determine the quantity of fuel available to the engine at sea level engine military rated power conditions utilizing both internal and external

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tanks by normal feed and alternate methods at the following aircraft attitudes. Fuel remaining in the system subsequent to initial flow fluctuations shall be considered unavailable at normal rated power.

(1) Normal ground attitude

(2) Take off attitude

(3) Level flight attitude (landing pattern conditions, low gross weight, low airspeed, low altitude)

(4) Landing attitude (touchdown)

(5) 10° greater angle than landing attitude

The location of the unavailable fuel shall be specified.

4.8.1.1.1 Internal tank switch-over and fuel transfer rates. Internal fuel tank manual switch-over provisions shall be demonstrated, as applicable. Internal fuel tank transfer rate capabilities shall be demonstrated under conditions considered most critical for the aircraft. Surge pressure resulting from feed and transfer flow fluctuation shall be measured and recorded.

4.8.1.2 Boost and transfer pump pressurization failure tests. Tests shall be conducted to determine the unavailable fuel quantities as a result of various fuel boost pump and transfer pump failures. If internal cell pressurization is used to supplement the boost pump supply system or if cell pressurization constitutes the primary supply system, testing to explore the effect of pressurization failure shall be conducted by disabling the pressurization system in conjunction with appropriate combinations of inoperative boost and transfer pumps, as applicable. The simulated engine(s) shall be operated at military power at normal aircraft flight and landing attitude to:

a. Determine the quantity of unavailable fuel with inoperative boost pumps which normally supply the engine

b. Determine the amount of fuel made unavailable with each fuel transfer pump made inoperative

c. Determine the amount of fuel made unavailable with significant combinations of fuel boost or transfer pumps made inoperative.

4.8.1.2.1 Vapor liquid ratio tests. With a fuel temperature of 110°F, and boost pumps inoperative, the vapor liquid ratio forming characteristics of the fuel system shall be determined for military and maximum engine power fuel flow. The vapor liquid ratio shall be measured at 2000-foot increments from sea level to an altitude which causes the vapor liquid ratio to be in excess of the actual capability of the

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engine fuel pump. The effects of airframe fuel filter bypassing or impending bypass on system vapor/liquid characteristics shall also be determined. Reid vapor pressure shall be determined for the fuel used during each simulated flight.

4.8.2 Aircraft fuel tank subsystem tests. Two or more internal fuel cells which are operationally integral through the use of interconnectors may be considered as one tank for the purpose of the tests herein.

4.8.2.1 Fuel tanks

4.8.2.1.1 Tank capacities. Capacity of each fuel tank shall be determined and compared with the design capacity approved by the procuring activity. The tanks shall be filled at the normal ground attitude.

4.8.2.1.2 Sump capacity. Sump capacity of each internal cell at normal ground attitude shall be obtained by measuring the fuel drained from the sump drain after all fuel possible has been pumped from the fuel system by the aircraft boost pumps. Determine the following:

a. That the capacity meets applicable design requirements

b. That all cells may be satisfactorily drained on the ground.

Residual fuel, if any, remaining after cells and sumps are drained shall be determined.

4.8.2.1.3 Expansion space. The aircraft shall be fueled to normal capacity, by pressure and gravity refueling, then measured fuel shall be added until the expansion space is filled. (Special provisions may be required.) This test shall be conducted to demonstrate that the internal fuel system is provided with adequate expansion space at the normal ground attitude without fuel spillage.

4.8.3 Fuel vent subsystem tests

4.8.3.1 Siphoning tests. Tests shall be conducted to determine the siphoning characteristics of the fuel vent system. After normal fueling, the aircraft shall be placed at normal flight attitude and the quantity of fuel, if any, discharged from the fuel vent opening shall be measured. After normal fueling at ground attitude, fuel shall be added until a steady stream of fuel is discharged from the fuel vent opening. The fuel being pumped into the system shall be shut off and the amount discharged from the vent opening, after shut off, shall be recorded.

4.8.3.2 Fuel tank purging. Purging tests shall be conducted in accordance with MIL-P-5902.

4.8.4 Fuel quantity gaging subsystem tests

4.8.4.1 Components. The components shall be tested in accordance with sampling plans A and B of MIL-G-26988.

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4.8.4.2 System tests. The system calibration tests shall be conducted in accordance with the requirements of MIL-G-7940. These tests shall not be conducted until approval has been granted by the procuring activity to the Engineering Report required in MIL-G-7940.

4.8.5 Ground fueling and defueling subsystem tests. The following test shall be conducted on the fuel system test rig to establish conformance with the design requirements.

4.8.5.1 Fueling. The time to fill the system shall be recorded using a starting condition as described in 3.12.1.2.b. The flow rates to the system shall be determined for various tank open conditions and for each tank open singly. Test shall be made for ground and aerial refueling nozzle inlet pressures of 20, 30, 40, 50, and 55 psig.

4.8.5.2 Surge pressure test. The refueling manifold shall be instrumented to determine surge pressure encountered during refueling. There shall be a minimum of one pressure pickup at each fuel level control valve, each ground and aerial refueling connection. The surge pressure shall be measured under the following conditions while refueling through each servicing refueling connection with a pressure of 50 ±5.

a. Maximum flow - all fuel control valves closed simultaneously

b. Maximum flow - fuel control valves closed singly

c. Maximum flow - all fuel control valves close as a result of filling to capacity.

4.8.5.3 Defueling test. Ground defueling shall be demonstrated. Aircraft internal pumps may be utilized to defuel the aircraft. The time required to defuel the internal cells shall be determined for the maximum discharge flow rate. The maximum flow and fuel remaining in the aircraft after defueling shall be recorded.

4.8.5.4 Fuel servicing nozzle clearance. If a switch or valve actuator has been installed on the adapter, a dummy fuel servicing nozzle incorporating the entire usable envelope allowed by MS29520 shall be fitted to the test rig to determine compliance with 3.12.1.1.3.

4.8.6 Fuel jettison system tests. Fuel jettisoning tests shall be performed to determine the average jettisoning rate and the quantity of fuel which cannot be jettisoned.

4.9 Ground and flight tests. Components shall be qualified or have passed approved safety of flight tests prior to conducting the ground and flight tests.

4.9.1 Fuel system icing. The airframe manufacturer shall demonstrate compliance with the requirements of 3.6.2.2.2. The icing test plan shall be approved by the procuring activity prior to testing.

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4.9.2 Engine feed and transfer subsystem tests

4.9.2.1 Center of gravity travel. Fuel system center of gravity travel tests shall be conducted for the airplane normal flight attitude with normal operation of engine fuel feed. The data shall illustrate the effects of all external fuel and auxiliary tank fuel as well as total internal fuel. The airplane shall be positioned at the normal ground attitude and pressure refueled to normal fuel capacity. The airplane shall then be positioned to the normal flight attitude and the reactions at scales under the main and nose gears recorded. Fuel shall be removed from the airplane through all engine feed lines at the normal maximum fuel flow rate down to unavailable fuel. During the fuel pump-out the reactions at the main and nose gears shall be recorded at 2-minute intervals. In analyzing the data the effects of landing gear position shall be accounted for.

4.9.2.2 Maximum altitude. The maximum altitude at which military power and maximum power operation can be maintained on suction feed shall be determined. The temperature of the fuel and a fuel sample for Reid vapor determinations shall be taken before and after the flights. The temperature of the fuel shall be as close as possible to 110°F immediately before take-off. The following tests shall be conducted:

a. A normal take-off and climb to minimum safe altitude shall be performed. Turn off all booster and transfer pumps. Multi-engine aircraft may test one engine at a time. Climb at military power at best climb speed to service ceiling or until 10 percent power loss occurs or until an objectional engine surge occurs.

b. Paragraph a. above shall be repeated using maximum power.

4.9.3 Tank system tests

4.9.3.1 Cell capacities, sump characteristics, and expansion space. The cell capacities, sump characteristics and expansion shall be determined on actual aircraft tanks in the same manner as determined in the fuel system rig tests.

4.9.4 Pressurization and fuel vent system tests. The vent system shall contain a pressure pick-up at each vent outlet, in each tank, and between each bladder cell and cell cavity. The pressure data and other data such as rate of climb, rate of dive, altitude, and quantity of fuel shall be recorded in sufficient quantity to demonstrate satisfactory operation of the vent system.

4.9.4.1 Fuel and vent system operational tests. Dyed fluid shall be used during the test conditions given below to mark any fuel impingement on the aircraft. The vent system shall be evaluated under the following conditions and maneuvers:

a. The aircraft shall be flown at maximum rate of climb to service ceiling followed by a maximum practical rate of descent dive minimum safe altitude. The dive shall be accomplished with the minimum safe quantity of fuel aboard

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b. The aircraft shall be flown at maximum engine power at service ceiling and at minimum safe altitude (approximately 5,000 feet) for 10 minutes duration each

c. Conduct a maximum speed run of maximum duration that is most conducive to aerodynamic heating of the fuel, fuel cell, and components

d. With full fuel tanks it shall be demonstrated that fuel spillage does not occur as a result of take-off and landing accelerations, afterburner and speedbrake application, and uncoordinated turns

e. Aircraft whose design performance includes inverted flight or negative 'G' shall be flown in this manner for a maximum period as determined by the model specification at the worst altitude condition.

4.9.4.2 Vent outlet icing tests. The vent outlets shall be observed following the aircraft icing tests to determine the extent of ice accumulation or blockage around the vent outlet. Any ice accumulation shall be photographed for submittal to the procuring activity.

4.9.4.3 Vent system failure tests. Failure of components of the fuel system shall be simulated to cause fuel to vent from the vent outlets. Any impingement of the fuel on the aircraft shall be recorded.

4.9.5 Inerting systems tests. Tests shall be conducted to investigate the most critical condition for which the inerting system was designed. Sufficient data shall be recorded to demonstrate satisfactory operation of the system.

4.9.6 Aerial refueling. The aerial refueling test program shall be conducted in the order losted.

4.9.6.1 Retrofit installations

4.9.6.1.1 Proximity flight test. A proximity flight test shall be conducted prior to hardware design and fabrication to determine feasibility, receptacle location and installation angle, or probe location and installation angle. The proximity test shall consist of the following and shall require color photo coverage.

a. Evaluate effects of tanker aerodynamic conditions and down wash on receiver during simulated hookups

b. Determine aerial refueling speed and altitude envelope and if afterburner and speed brakes, trailing and leading edge flaps are required for altitude and speed envelope

c. Instrument the receiver aircraft to measure the fuselage pitch angle while in position behind tanker during combinations of the following conditions:

(1) Heavy gross weight

(2) Light gross weight

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(3) Aerial refueling speed and altitude envelope as determined in item b above.

(4) Employ modulated afterburner range as established in item b above, if applicable.

d. Determine suitability of receptacle or probe location by making simulated hookups.

4.9.6.1.2 Retrofit installation ground test. The retrofit installation shall be tested in accordance with 4.8.5.1, 4.8.5.2, and 4.9.2.1.

4.9.6.1.3 Retrofit installation final testing. Retrofit installation final testing shall be the same as the production installation testing.

4.9.6.2 Production installations

4.9.6.2.1 Aerial refueling ground tests. The following ground tests shall be conducted prior to the first aerial refueling flight test:

a. Conduct a structural test to ultimate load for design conditions specified in MIL-A-8865, for the receptacle support structure or probe support structure as installed on the aircraft.

b. Perform a complete functional test of all installed aerial refueling components.

c. Slipway door/receptacle or probe and probe doors shall demonstrate ability to extend and retract during hydraulic actuation at -65°F temperatures and icing conditions, if not previously demonstrated during the development phase. Failed open provisions shall also be demonstrated during above temperature and icing conditions. Time of opening and hydraulic pressures shall be recorded. The actuation shall be accomplished under simulated flight air loads for the most adverse conditions.

d. Drains shall demonstrate the ability to remove rain and fuel spillage during all appropriate ground and flight attitudes

e. (Receptacle installations only.) Evaluate the slipway and transion surfaces from the slipway to the receptacle for possible boom nozzle hang-up point and for proper location of the night refueling lights.

f. (Probe installation only.) Evaluate the probe for possible drogue hang-up points and for proper location of the night refueling lights.

4.9.6.2.2 Flight test. The tests outlined below shall be conducted as a part of Category I testing. Photo coverage is required for all test flights. Items a and b shall be evaluated for both light and heavy gross weight configurations throughout the speed envelope. The fuselage pitch angle for the receiver aircraft during refueling shall be recorded for both light and heavy weight conditions.

a. (Receptacle installations only.) Explore entire tanker boom envelope including elevation extension and azimuth in 5-degree and 6-foot increments for disconnects ability, slipway interferences, and the ability of the tanker boom to tow the receiver.

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b. (Probe installations only.) Explore the entire tanker hose envelope for possible hose interference and for disconnect ability.

c. Determine the optimum receiver refueling envelope in relation to the following items:

(1) Altitude and speed

(2) Boom extension, azimuth and elevation angles (as applicable)

(3) Wing flap configuration

(4) Use of modulated afterburner and speed brakes, if required for aerial refueling.

d. Determine effect of light to moderate air turbulence and moderate to heavy air turbulence on the time required to accomplish hook-up.

e. Determine adequacy of indicator lights, tanker directional lights, cockpit switch and control locations.

f. Evaluate the night lighting adequacy during a transition from twilight to dusk to full dark conditions. The use of the rheostat and any resulting blinding or objectional reflections shall be recorded.

g. Aerial refuel the receiver to the maximum aircraft gross weight required by mission definition. A minimum of five hook-ups shall be made with the receiver having a weight of 90 percent of maximum gross weight. The tanker shall record the number of refueling pumps and corresponding flow rates during the refueling. The receiver shall attempt different valve shut off combinations including "gang" closing at maximum fuel flow.

h. (Receptacle installations only.) The following types of disconnects shall be accomplished:

(1) Manifold pressure switch

(2) Receiver pilot initiated by both normal amplifier system and amplifier override

(3) Tanker boom operator initiated by both normal and amplifier override

(4) All tanker boom automatic envelope disconnects including combinations of left and right azimuth, elevation extremes, and minimum and maximum boom extensions.

i. (Receptacle installations only.) Evaluate emergency refueling such as override features, stiff boom, fail safe features. Two brute-force-disconnects shall be demonstrated; each performed after a 1-hour flight at 40,000 feet altitude.

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j. (Receptacle installations only.) When reverse aerial refueling is required reverse flow tests shall be conducted and shall consist of the receiver pumping fuel to the tanker at predetermined flow rates and pump combinations. A disconnect shall be accomplished during maximum fuel flow conditions.

4.9.7 Fuel jettison system

4.9.7.1 Fuel impingement. Demonstrate that no fuel or fumes impinge or enter any portion of the aircraft or cause any other unsafe conditions under the following conditions (flaps and gear up) at design gross weight. Dumping rate shall be recorded during the following operating condition:

a. Climb at the one engine inoperative best rate-of-climb, with the critical engine inoperative and the remaining engines at maximum continuous power. The critical engine defined as that engine which, when inoperative, will cause flight conditions such as yaw to be most unfavorable for safe fuel jettisoning

b. Level flight at 1.4 times the indicated stalling speed, if the results of the previous tests show that this condition could be critical.

4.9.7.2 Controllability. A maximum quantity of fuel shall be jettisoned to determine any adverse effects on the controllability of the aircraft.

5. PREPARATION FOR DELIVERY

5.1 Not applicable

6. NOTES

6.1 Intended use. This specification is intended for use by aircraft manufacturers in designing and testing fuel systems for air breathing engine aircraft.

6.2 Definitions

a. Available Fuel - Fuel which can be pumped from the aircraft when the tanks have been filled to automatic shut-off at the normal ground refueling attitude subsequent to initial flow fluctuations.

b. Unavailable Fuel - Fuel remaining in the aircraft after the available fuel has been removed.

c. Sump Fuel - Fuel which can be drained from the aircraft sumps after the available fuel has been removed from the aircraft.

d. Trapped Fuel - Fuel remaining in the aircraft after the sump fuel has been drained.

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e. Slipway - The specially designed "beefed-up" fuselage skin area forward of the receptacle which serves as a guide to assist the boom operator in hook-up operations. The use of this area is required to stabilize the boom nozzle in the vertical and horizontal planes, due to the relative motions which exist prior to contact and hook-up. Relative motions which exist between the receiver and boom are normally produced; when the aircraft are subjected to weather turbulence, boom hook-ups attempted at lower boom elevations (35° to 40°), out of rig booms, and inherent boom instability.

f. Brute-Force-Disconnect - The process of disconnecting against the force of locked receptacle toggle latches.

g. Receiver Director Lights - Two rows of lights are contained on the bottom fuselage forward of the wheel well of the KC-135 tanker. These lights give location information to direct the receiver aircraft pilot in the forward-aft and up-down boom envelope.

h. Reverse Refueling - The process of transferring fuel from the receiver to the tanker.

Custodian:
Air Force - 11
Review activities:
Air Force - 11, 80, 82

Preparing activity:
Air Force - 11

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APPENDIX

10. SCOPE

10.1 This appendix lists the data necessary for the procuring activity to properly evaluate a developmental fuel system.

20. APPLICABLE DOCUMENTS

20.1 The following specification of the issue in effect on the date of invitation for bids or request for proposal, forms a part of this data list to the extent specified herein:

MIL-F-38363 Fuel System, Aircraft Design, Performance Installation, Testing and Data Requirements, General Specification For

30. Requirements. The following data/reports are necessary to thoroughly define the fuel system for the procuring activity.

a. Engineering Design Analysis Report. This report shall include the following:

(1) Result of trade studies performed to arrive at the basic fuel system configuration

(2) Temperature calculation showing maximum fuel temperature resulting from aero/thermodynamic heating of the fuel and maximum ambient air temperature

(3) Calculated system performance

(4) Predicted problem areas with proposed method of solution

(5) Development test program

(6) Special facilities required for testing of fuel system

(7) Deviations to MIL-F-38363 and reason for deviation.

b. Fuel System Functional Analysis Report. This report shall include the following:

(1) Description of fuel system

(2) Fuel system schematic diagrams identifying all components, lines, sizes and relative location

(3) List of QPL components giving name and location of component, manufacturer's part number, vendor's part number

(4) List of off-the-shelf components giving name and location of component, manufacturer's part number, vendor part number, applicable military specification, approval status, procurement or model specification, past usage

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(5) List of critical components giving name and location of component, manufacturer's part number, vendor part number, detail drawings, applicable military specification, procurement specification, and test report

(6) Fuel system wiring schematics from distribution bus to all fuel system components

(7) Perspective or isometric drawing illustrating all major components and fuel lines in relation to the airframe structure

(8) Outline of fuel scheduling and management

(9) Failure analysis describing the effect of each possible single failure on the operation of the system or subsystem

(10) Fuel system passive defense analysis

(11) Procuring activity approval items required by paragraphs of MIL-F-38363:

(a) 3.6.2.5.1.4

(b) 3.7.8.1

(c) 3.13.1.1

(d) 3.13.1.2

(e) 3.13.1.3.1

(f) 3.13.1.3.2

(g) 3.13.2

(h) 4.5

(i) 4.7.1

(j) 4.7.2

(k) 4.9

(l) 4.9.1

c. Systems Test Plan. Outline proposed test program.

d. Systems Test Report. Submit results of system tests required in 4.8 and 4.9 of MIL-F-38363, and motion pictures of aerial refueling flight tests.

e. Fuel System Detailed Weight Statement. The following weight statement shall be submitted as complete as possible during the proposal stage. The weight statement should be up-dated for design reviews as data becomes available.

Fuel System Detailed Weight Statement:

ITEM	WEIGHT
(1) Tanks	_____
(2) Integral Tank Seals and Sealant	_____

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(3) Backing Board _____

(4) Tank Supports and Padding _____

(5) Tank Bay Sealing _____

(6) Tank Release and Controls _____

(7) Fuel Quantity Gaging System _____

(8) Fuel Heating and Controls _____

(9) Fuel Vapor Recovery _____

(10) Fuel Vapor Inerting System and Suppression

Cycle and Supports _____

Generators _____

Controls, Etc. _____

(11) Pump Installation

Engine Driven _____

Booster _____

Hand (Including Controls) _____

Transfer

Electrical System (Including Controls) _____

Pump Supports _____

(12) Ground Fueling System

Plumbing _____

Controls _____

(13) Aerial Refueling Hardware _____

Plumbing _____

Controls _____

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Additional Structure _____

(14) Engine Feed and Transfer System _____

Plumbing _____

Controls _____

(15) Vent System _____

(16) Pressurization System _____

(17) Jettison System _____

(18) Fuel Drain System _____

TOTAL _____

f. System Detail Design Data. The applicable portion of the following list shall be submitted as complete as possible during the proposal stage. The list should be updated for design reviews as data becomes available.

(1) Fuel: Primary_____Alternate_____

Consumption at takeoff_____(lb/hr/engine)

Consumption at cruise_____(lb/hr/engine)

(2) Tanks, Fixed: (Integral, Bladder, Non-Integral, External)

Specification_____

Type_____

Number Quantity/Tanks_____

Location_____(Station/WL/BL)

Quantity_____

Design Crash_____(g's/capacity/angle)

Burst_____(psig)

Negative Pressure Capability_____(psig)

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Overall Dimensions _____ (in)

Largest Opening/Location _____ (in/sta)

Sump Location/Capacity _____ (sta/gal)

Expansion Space _____ (gal)

Drain Location _____

Baffles Axial/Transverse _____ (Location and Number)

Supplier _____

(3) Fuel Capacity, Fixed:

Total Fixed Capacity _____ (gal)

Trapped Main Fuel _____ (gal)

Usable Main Fuel _____ (gal)

Sump Fuel _____ (qt)

Expansion Space _____ (gal)

(4) Tanks, Auxiliary: (Internal, External) Same as (2).

(5) Fuel Capacity, Auxiliary:

Total Auxiliary Capacity _____ (gal)

Trapped Auxiliary Fuel _____ (gal)

Usable Auxiliary Fuel _____ (gal)

Sump Fuel _____ (gal)

Expansion Space (Auxiliary) _____ (gal)

(6) Tank Materials:

Integral Structure _____

Specification _____

Heat-Treat/Grade _____

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Sealants

Specification _____

Method of Application _____

Bladder Support Structure _____

Bladder, Self-Seal

Specification _____

Description _____

Bladder, Non-Seal

Specification _____

Description _____

Non-Integral Internal Tank Structure

Specification _____

Heat-Treat/Grade _____

Ultimate Tensile Strength _____

(7) Refuel:

Gravity Fill

Drain Line Size _____ (in)

Drain Discharge Point _____

Filler Locations _____

Maximum Fill Rate _____ (gpm)

Single Point

Location _____

Receptacle Type/Std _____

Rated Flow/Pressure _____ (pph/psig)

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Level Control Valve Pressure Check Method _____

Line Size _____ (in)

Burst Pressure _____ (psig)

Fill Sequence _____

Aerial Fill

Method _____

Location _____ (sta/WL/BL)

Rated Flow/Pressure _____ (gal/psig)

Line Size _____ (in)

Fill Sequence _____

(8) Defuel:

Fixed Tanks Defuel Subsystem

Description _____

Location _____ (sta/WL/BL)

Receptacle Type/Std _____

Line Size _____ (in)

Rated Flow/Pressure _____ (gpm/psig)

Auxiliary Tanks Defuel Subsystem

Description _____

Location _____ (sta/WL/BL)

Receptacle Type/Std _____

Line Size _____

Rated Flow/Pressure _____ (gpm/psig)

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(9) Jettison:

Description _____

Location _____ (sta/WL/BL)

Line Size _____ (in)

Rated Flow/Pressure _____ (gpm/psig)

Pressure Source _____

Method of Low Quantity Cut-Off _____

(10) Vents:

Anti-Syphon Design Description _____

Outlet Locations _____ (sta/WL/BL)

Size of Line _____ (in)

Lightning Protection _____

Special Design Feature _____

(11) Negative g Provisions:

Type Description _____

Time at Military Power _____ (min-sec)

Time at Cruise Power _____ (min-sec)

(12) C.G. Control:

Tank Usage Sequence _____

Normal C.G. Travel _____

Abnormal C.G. Travel Possibilities _____

Location and Type of Fuel Management Controls _____

(13) Critical Components:

Number and Description of Surveillance Items _____

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Number and Description of Single Failure Critical Items_____

(14) Boost Pumps:

Location_____(sta/WL/BL)

Power_____

Circuit Protection_____

Flow/Pressure at TO Power_____

Flow/Pressure at Cruise_____

Type_____

Speed_____

Supplier_____

Special Features_____

(15) Other Pumps: (Transfer Jettison, Ejection, etc.)

Function_____

Type of Pump_____

Location_____

Power_____

Circuit Protection_____

Speed_____

Flow/Pressure in Operation_____(gpm-psig)

(16) Strainers:

Type_____

Overall Dimensions_____

Location_____

Inlet Size_____

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Outlet Size_____

P at TO Flow/60°F_____

P at By-Pass/60°F_____

Heater Power_____

Mesh or Micron Rating_____

Supplier_____

(17) Plumbing:

Hose Specification_____

Fitting Material_____

Flex Coupling Type _____

Rigid Line Material_____

Wall Thickness_____

Ultimate Tensile Strength_____

Design Burst Factor_____

Fitting Type_____

Line Sizes for:

Vents_____

Crossfeed_____

Jettison_____

Single Point Refueling_____

(18) Quantity Instrumentation:

Type_____

Overall Indicator Size_____

Switch Positions_____

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Probe Location_____

Low Level Locations_____

Low Level Quantity_____

Supplier_____

(19) Flow Instrumentation:_____

Type_____

Sensor Location_____

Indicator Ranges_____

Accuracy_____

Readout Data_____

Supplier_____

(20) Pressure Instrumentation:

Type_____

Function_____

Sensor Location_____

Indicator Ranges_____

Actuation Pressure_____

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